Data Collection and Preprocessing Phase

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	6 Marks

Data Exploration and Preprocessing Template

Source Name	Description	Format	Size
Kaggle – Online Payments Fraud Detection Dataset	Financial transaction dataset containing labeled fraud instances for supervised learning.	CSV	~6.3 Million records

Missing Values Analysis

Issue Identified	Observation	Resolution Plan
Missing Values	No significant missing values detected in core features	No imputation required

Action Taken:

- Verified using `isnull().sum()` and `.info(show_counts=True)` functions.
- Confirmed dataset completeness.

Class Imbalance

Issue Identified	Observation	Resolution Plan
Severe Class Imbalance	Fraud cases ≈ 0.13% of total data	Focus on Recall metric; avoid accuracy bias

Outlier Detection

- Visuals created through boxplots
- Fraud transactions show significantly higher transaction amounts.
- Extreme value transactions retained (important fraud indicators).
- No outlier removal applied to avoid information loss.

Section	Description

Data Overview	**Metric** — **Value**
	Total Records — ~6.3 Million
	Total Features — 8
	Target Variable — `isFraud` (Binary)
	Data Type — Structured Tabular
	Source — Hosted on **Kaggle**
	
	Structural Observations
	
	• No missing values in critical features.
	• Numerical and categorical feature mix.
	• Highly imbalanced fraud distribution (~0.13% fraud cases).
	• Suitable for supervised binary classification.
	
	Basic Statistics (Key Numerical Features)
	
	• **Transaction Amount**
	  ○ Mean: Moderate
	  ○ Median: Lower than mean (right-skewed distribution)
	  ○ Max: Extremely high (indicative of potential fraud spikes)
	• **Account Balances**
	  ○ Wide variance observed.
	  ○ Several zero-balance transitions detected.

Univariate Analysis	Measures used: • Mean • Median • Standard Deviation • Frequency Distribution • Value Counts

Bivariate Analysis	**Key Relationships Identified**
	
	Amount vs Fraud
	
	• Fraudulent transactions show significantly higher average transaction amount.
	
	Transaction Type vs Fraud
	
	• Majority of fraud cases occur in TRANSFER transactions.
	
	Sender Balance vs Fraud
	
	• Fraud often occurs when sender's balance is fully depleted.
	
	Correlation Analysis
	
	• Strong correlation between:
	  ○ oldbalanceOrg & newbalanceOrig
	  ○ oldbalanceDest & newbalanceDest
	• Weak direct correlation between amount and fraud (non-linear relationship).
	
	Techniques Used:
	
	• Correlation Matrix & Heatmap
	• Box Plots
	• Grouped Aggregation

<table>
<tr><td>Multivariate Analysis</td><td>

Observations

- Fraud transactions often show:
  - High transaction amount
  - Zero sender balance after transaction
  - Transfer transaction type
- Combined feature behavior improves fraud separability.
- Interaction between amount and transaction type strengthens fraud prediction.

Advanced Insights:

- Tree-based models capture nonlinear interactions effectively.
- Feature importance indicates:
  - `amount`
  - `type`
  - `oldbalanceOrg`

  as dominant predictors.

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<tr><td>Outliers and Anomalies</td><td>

Description

Identification and handling of extreme or abnormal values.

Findings

- Extreme high-value transactions detected.
- Sudden full-balance depletion transactions observed.
- These are not data errors but meaningful fraud signals.

Treatment Strategy

Issue	Action Taken
Extreme transaction amounts	Retained (fraud indicator)
Balance inconsistencies	Validated; retained
Synthetic anomalies	No removal to avoid loss of predictive signal

No outlier removal was performed since extreme values contribute to fraud detection performance.

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<tr><td colspan="2">Data Preprocessing Code Screenshots</td></tr>
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Loading Data

```
import os
pip install kaggle
!kaggle datasets download -d rupakroy/online-payments-fraud-detection-dataset
!unzip online-payments-fraud-detection-dataset.zip
df_orig=pd.read_csv('PS_20174392719_1491204439457_log.csv')
df_orig
```

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest	oldbalanceDest	newbalanceDest	isFraud	isFlaggedFraud
0	1	PAYMENT	9839.64	C1231006815	170136.00	160296.36	M1979787155	0.00	0.00	0	0
1	1	PAYMENT	1864.28	C1666544295	21249.00	19384.72	M2044282225	0.00	0.00	0	0
2	1	TRANSFER	181.00	C1305486145	181.00	0.00	C553264065	0.00	0.00	1	0
3	1	CASH_OUT	181.00	C840083671	181.00	0.00	C38997010	21182.00	0.00	1	0
4	1	PAYMENT	11668.14	C2048537720	41554.00	29885.86	M1230701703	0.00	0.00	0	0

Handling Missing Data

```
df.info(show_counts=True)
df.isnull().sum()
df.info(show_counts=True)
```

```
df.info(show_counts=True)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6362620 entries, 0 to 6362619
Data columns (total 8 columns):
 #   Column          Non-Null Count    Dtype
---  ------          --------------    -----
 0   step            6362620 non-null  int64
 1   type            6362620 non-null  object
 2   amount          6362620 non-null  float64
 3   oldbalanceOrg   6362620 non-null  float64
 4   newbalanceOrig  6362620 non-null  float64
 5   oldbalanceDest  6362620 non-null  float64
 6   newbalanceDest  6362620 non-null  float64
 7   isFraud         6362620 non-null  object
dtypes: float64(5), int64(1), object(2)
memory usage: 388.3+ MB
```

Data Transformation

```
df.drop(['isFlaggedFraud', 'nameOrig', 'nameDest'], axis=1, inplace=True)

#Encoding the string values in Type column to integer mapping
le=LabelEncoder()
df['type']=le.fit_transform(df['type'])

#Extracting the map inside of Label Encoder
mapping=dict(zip(le.classes_, range(len(le.classes_))))
print(mapping)
{'CASH_IN': 0, 'CASH_OUT': 1, 'DEBIT': 2, 'PAYMENT': 3, 'TRANSFER': 4}

scaler=StandardScaler()
x_scaled=scaler.fit_transform(x)
x_scaled
```

Feature Engineering	`df.loc[df['isFraud']==0, 'isFraud']='Legit Transaction'` `df.loc[df['isFraud']==1, 'isFraud']='Fraud Transaction'`
Save Processed Data	`df=df_orig.copy()` `df`